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ABNORMALITY DIAGNOSIS SYSTEM

Abstract

An abnormality diagnosis system includes a storage and processing circuitry. The storage stores cluster data. The cluster data indicates a distribution of clusters by type of abnormality in a hydraulic pressure control device. The processing circuitry generates evaluation data on features in hydraulic pressure recorded while operating the hydraulic pressure control device to be evaluated in a diagnosis pattern. The processing circuitry determines whether an abnormality has occurred in the hydraulic pressure control device based on a result of determining which cluster in the cluster data each of the features at multiple time points in the evaluation data belongs to.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-035848, filed on Mar. 8, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to an abnormality diagnosis system for a hydraulic pressure control device.

2. Description of Related Art

[0003] Japanese Laid-Open Patent Publication No. 2021-085480 discloses an abnormality diagnosis system for identifying the cause of an abnormality in shift control of an automatic transmission. The abnormality diagnosis system determines the cause of the abnormality in the automatic transmission based on the aspects of transient changes in a rotational speed of the automatic transmission.

[0004] In a case where the determination is performed by observing the fluctuation of the rotational speed in the automatic transmission, there is a possibility that the cause of the abnormality cannot be identified. For example, it is difficult to determine a failure of a component in the hydraulic pressure control device of the automatic transmission from the fluctuation of the rotational speed.

SUMMARY

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0006] An abnormality diagnosis system according to an aspect of the present disclosure includes processing circuitry and a storage. The storage of the abnormality diagnosis system stores cluster data. The cluster data indicates a distribution of clusters data by type of abnormality for features at multiple time points. The cluster data is generated by acquiring, a number of times, data on the features for fluctuation of hydraulic pressure obtained when operating, in a predetermined diagnosis pattern, a hydraulic pressure control device with an identified abnormality status and an identified type of abnormality. The processing circuitry of the abnormality diagnosis system is configured to generate, as evaluation data, data on the features in hydraulic pressure that is recorded while operating the hydraulic pressure control device to be evaluated in the diagnosis pattern. The processing circuitry of the abnormality diagnosis system is configured to determine whether an abnormality has occurred in the hydraulic pressure control device based on a result of determining which cluster in the cluster data each of the features at multiple time points in the evaluation data belongs to by comparing the features at the time points included in the evaluation data with the cluster data.

[0007] The decision boundary for determining which cluster the feature in the evaluation data belongs to may be determined using a machine learning algorithm such as a support vector machine.

[0008] The abnormality diagnosis system performs a determination using the data on the fluctuation of the hydraulic pressure of the hydraulic pressure control device. This allows the abnormality diagnosis system to acquire changes in the behavior of a hydraulic pressure resulting from an abnormality and determine that an abnormality has occurred in the hydraulic pressure control device.

[0009] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a schematic diagram showing an embodiment of an abnormality diagnosis system and a hydraulic pressure control device to be diagnosed.

[0011] FIG. 2 is a schematic diagram showing a configuration of the hydraulic pressure control device.

[0012] FIG. 3 is a graph showing an example of a diagnosis pattern.

[0013] FIG. 4 is a graph showing an example of clustering.

[0014] FIG. 5 is a flowchart showing the flow of processes executed by the abnormality diagnosis system.

[0015] FIG. 6 is a table showing the result of clustering.

[0016] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0017] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order.

[0018] Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0019] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0020] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

[0021] Hereinafter, an embodiment of the abnormality diagnosis system **100** will be described with reference to FIGS. 1 to 6.

Configuration of Abnormality Diagnosis System **100**

[0022] FIG. 1 shows the abnormality diagnosis system **100** and an automatic transmission **200** including a hydraulic pressure control device **240** to be diagnosed. When diagnosing the hydraulic pressure control device **240**, as shown in FIG. 1, the abnormality diagnosis system **100** is connected to the automatic transmission **200**. The abnormality diagnosis system **100** diagnoses the hydraulic pressure control device **240** by operating the hydraulic pressure control device **240** in a predetermined diagnosis pattern.

[0023] As shown in FIG. 1, the abnormality diagnosis system **100** includes processing circuitry **110**, a storage **120**, and a display **130**. The processing circuitry **110** includes a CPU that executes processing in accordance with a program, and a ROM in which the program is stored. The storage **120** stores data. The abnormality diagnosis system **100** is, for example, a personal computer or a workstation.

Configuration of Automatic Transmission **200**

[0024] The automatic transmission **200** including the hydraulic pressure control device **240** to be inspected is mounted on, for example, an automobile. In FIG. 1, a continuously variable transmission is shown as an example of the automatic transmission **200**. The automatic

transmission **200** includes a first pulley **210**, a second pulley **220**, and a belt **230** wound around the first pulley **210** and the second pulley **220**. The first pulley **210** is connected to an input shaft **211**. The second pulley **220** is connected to an output shaft **221**. The automatic transmission **200** transmits a driving force via a belt **230**. In the automatic transmission **200**, the hydraulic pressure control device **240** controls a first hydraulic pressure P_p supplied to the first pulley **210** and a second hydraulic pressure P_s supplied to the second pulley **220**. Accordingly, the winding radius of the belt **230** on each of the first pulley **210** and the second pulley **220** changes. That is, the hydraulic pressure control device **240** changes the gear ratio of the automatic transmission **200** by changing the winding radius of the belt **230**.

[0025] When diagnosing the hydraulic pressure control device **240**, a drive motor **300** is connected to the input shaft **211**, and a load motor **400** is connected to the output shaft **221**. Hydraulic pressure sensors **500** that detect the hydraulic pressure of each part are connected to the hydraulic pressure control device **240**. The diagnosis of the hydraulic pressure control device **240** is performed, for example, as a pre-shipment inspection after the automatic transmission **200** is manufactured. The drive motor **300** reproduces an input of the driving force to the automatic transmission **200** from a driving power source for an automobile. The load motor **400** reproduces a load acting on the automatic transmission **200**.

Configuration of Hydraulic Pressure Control Device **240**

[0026] As shown in FIG. **2**, the hydraulic pressure control device **240** includes a regulator valve **241**, a modulator valve **242**, a first control valve **247**, and a second control valve **248**. The hydraulic pressure control device **240** includes a first linear solenoid valve **243**, a second linear solenoid valve **244**, a first hydraulic damper **245**, and a second hydraulic damper **246**.

[0027] The regulator valve **241** regulates the hydraulic pressure of the oil supplied from the oil pump. The hydraulic pressure regulated by the regulator valve **241** is a line pressure PL . The oil regulated to the line pressure PL is supplied to the first control valve **247** and the second control valve **248**. The hydraulic pressure adjusted to the line pressure PL is also supplied to the modulator valve **242**.

[0028] The first control valve **247** supplies the oil regulated to the line pressure PL to the first pulley **210**. The first control valve **247** controls the amount of oil supplied to the first pulley **210**. The first control valve **247** increases the first hydraulic pressure P_p by supplying the oil regulated to the line pressure PL to the first pulley **210**. The first control valve **247** controls the amount of oil discharged from the first pulley **210**. The first control valve **247** lowers the first hydraulic pressure P_p by discharging oil from the first pulley **210**.

[0029] The second control valve **248** supplies the oil regulated to the line pressure PL to the second pulley **220**. The second control valve **248** controls the amount of oil to be supplied to the second pulley **220**. The second control valve **248** increases the second hydraulic pressure P_s by supplying the oil regulated to the line pressure PL to the second pulley **220**. The second control valve **248** also controls the amount of oil discharged from the second pulley **220**. The second control valve **248** lowers the second hydraulic pressure P_s by discharging oil from the second pulley **220**.

[0030] When the first hydraulic pressure P_p is increased and the second hydraulic pressure P_s is decreased, the winding radius of the belt **230** around the first pulley **210** increases and the winding radius of the belt **230** around the second pulley **220** decreases. When the second hydraulic pressure P_s is increased and the first hydraulic pressure P_p is decreased, the winding radius of the belt **230** around the second pulley **220** increases and the winding radius of the belt **230** around the first pulley **210** decreases. In this manner, the hydraulic pressure control device **240** changes the gear ratio in the automatic transmission **200**.

[0031] The first control valve **247** and the second control valve **248** are each controlled by a signal hydraulic pressure. The modulator valve **242** regulates the line pressure PL to the signal hydraulic pressure. The oil adjusted to the signal hydraulic pressure is supplied to the first linear solenoid valve **243** and the second linear solenoid valve **244**.

[0032] The first linear solenoid valve **243** controls the signal hydraulic pressure supplied to the first control valve **247**. The second linear solenoid valve **244** controls the signal hydraulic pressure supplied to the second control valve **248**. The signal hydraulic pressure supplied to the first control valve **247** is controlled by opening and closing the first linear solenoid valve **243**. The signal hydraulic pressure supplied to the second control valve **248** is controlled by opening and closing the second linear solenoid valve **244**.

[0033] The first hydraulic damper **245** is provided in an oil passage connecting the first linear solenoid valve **243** to the first control valve **247**. The second hydraulic damper **246** is provided in an oil passage connecting the second linear solenoid valve **244** to the second control valve **248**.

Cluster Data

[0034] Cluster data is stored in the storage **120** of the abnormality diagnosis system **100**. The cluster data is generated by acquiring, a number of times, the data on the features for the fluctuation of hydraulic pressure obtained when operating, in a predetermined diagnosis pattern, the hydraulic pressure control device **240** with an identified abnormality status and an identified type of abnormality. The cluster data indicates the distribution of clusters by type of abnormality for each feature at multiple times points in the predetermined diagnosis pattern. To generate the reference cluster data, the hydraulic pressure is measured while operating the hydraulic pressure control device **240** in the predetermined diagnosis pattern. For example, the line pressure PL, the first hydraulic pressure Pp, and the second hydraulic pressure Ps among the hydraulic pressures at multiple parts in the hydraulic pressure control device **240** are measured.

[0035] FIG. **3** shows changes in a target hydraulic pressure Ppt of the first hydraulic pressure Pp in the predetermined diagnosis pattern. The predetermined diagnosis pattern is defined by a target hydraulic pressure PLt of the line pressure PL, the target hydraulic pressure Ppt of the first hydraulic pressure Pp, and a target hydraulic pressure Pst of the second hydraulic pressure Ps. The predetermined diagnosis pattern is defined such that the line pressure PL, the first hydraulic pressure Pp, and the second hydraulic pressure Ps are fluctuated over a predetermined period so as to diagnose an abnormality in each portion of the hydraulic pressure control device **240** from the changes in the measured hydraulic pressure. The target hydraulic pressure PLt of the line pressure PL, the target hydraulic pressure Ppt of the first hydraulic pressure Pp, and the target hydraulic pressure Pst of the second hydraulic pressure Ps in the predetermined diagnosis pattern have different fluctuation patterns.

[0036] The actual hydraulic pressure varies with a delay with respect to the fluctuation of the target hydraulic pressure. The actual hydraulic pressure may overshoot the target hydraulic pressure. Instead of immediately converging to the target hydraulic pressure, the actual hydraulic pressure may oscillate up and down across the target hydraulic pressure until it converges to the target hydraulic pressure. The data indicating the changes in differential pressure between the target hydraulic pressure and the measured hydraulic pressure includes information on the delay, overshoot, and oscillation of the actual hydraulic pressure with respect to the target hydraulic pressure. Such delay, overshoot, and oscillation occur even in the hydraulic pressure control device **240** in which no abnormality has occurred.

[0037] In the predetermined diagnosis pattern, a time is divided into periods in accordance with the type of abnormality to be determined, and in each time period, the hydraulic pressures at multiple parts controlled by the hydraulic pressure control device **240** are controlled in a pattern suitable for determining the corresponding abnormality. The cluster data is generated by repeatedly measuring the line pressure PL, the first hydraulic pressure Pp, and the second hydraulic pressure Ps a number of times while operating the hydraulic pressure control device **240** in the predetermined diagnosis pattern. The cluster data is generated using the hydraulic pressure control device **240** in which the abnormality to be determined has occurred, in addition to the hydraulic pressure control device **240** in which no abnormality has occurred.

[0038] For example, each hydraulic pressure is measured while the hydraulic pressure control

device **240** in which no abnormality has occurred is operated in the predetermined diagnosis pattern. By performing this measurement a predetermined number of times, the data on the cluster **C_0** in a state where no abnormality has occurred is acquired. The data on the cluster **C_1** in the state where the abnormality has occurred in the first hydraulic damper **245** is acquired by similarly performing the measurement the predetermined number of times using the hydraulic pressure control device **240** in which the abnormality has occurred in the first hydraulic damper **245**. The data on the cluster **C_2** in the state where the abnormality has occurred in the second hydraulic damper **246** is acquired by similarly performing the measurement the predetermined number of times using the hydraulic pressure control device **240** in which the abnormality has occurred in the second hydraulic damper **246**. The data on the cluster **C_3** in the state where the abnormality has occurred in the first control valve **247** is acquired by similarly performing the measurement the predetermined number of times using the hydraulic pressure control device **240** in which the abnormality has occurred in the first control valve **247**. The data on the cluster **C_4** in the state where the abnormality has occurred in the second control valve **248** is acquired by similarly performing the measurement the predetermined number of times using the hydraulic pressure control device **240** in which the abnormality has occurred in the second control valve **248**.

[0039] The cluster data is obtained by collecting the data on the features of fluctuation for each of the line pressure **PL**, the first hydraulic pressure **Pp**, and the second hydraulic pressure **Ps** so as to indicate the distribution for each type of abnormality. Examples of the features include an overshoot amount of the hydraulic pressure, a time constant indicating a response delay of the hydraulic pressure, the amplitude in oscillation of the hydraulic pressure, the frequency of the oscillation of the hydraulic pressure, and a damping coefficient of the oscillation of the hydraulic pressure. The cluster data is data in which the distribution of the features at multiple time points in the predetermined diagnosis pattern is classified for an abnormality status and the type of abnormality based on the data on the measured hydraulic pressure so as to indicate the distribution.

[0040] FIG. **4** is a part of the cluster data for the first hydraulic pressure **Pp**, showing the cluster distribution for the overshoot amount at a certain time point. In the example illustrated in FIG. **4**, three clusters are illustrated in a one-dimensional coordinate system in which the overshoot amount is an explanatory variable. In FIG. **4**, each data point in the overshoot amount of the cluster **C_0** in a state where no abnormality has occurred is indicated by a blank circle. Each data point in the overshoot amount of the cluster **C_1** in a state where an abnormality has occurred in the first hydraulic damper **245** is indicated by a blank triangle. Each data point in the overshoot amount of the cluster **C_3** in a state where an abnormality has occurred in the first control valve **247** is indicated by a blank square. In FIG. **4**, the centroid of each cluster is indicated by a cross mark.

[0041] The cluster data is data in which the distribution of clusters for each type of abnormality at multiple time points is collected for the features of the hydraulic pressures in the predetermined diagnosis pattern.

[0042] When the hydraulic pressure control device **240** is operated in the same diagnosis pattern, the hydraulic pressure varies depending on the abnormality status and the type of abnormality. Therefore, as shown in FIG. **4**, the distribution of data in each feature varies depending on the abnormality status and the type of abnormality. Thus, the abnormality diagnosis system **100** generates, as evaluation data, data on the features in hydraulic pressure that is recorded while operating the hydraulic pressure control device **240** to be evaluated in the predetermined diagnosis pattern. Then, the abnormality diagnosis system **100** compares each of the features at multiple time points included in the evaluation data with the cluster data. In this manner, the abnormality diagnosis system **100** performs clustering to determine which cluster in the cluster data each of the features at multiple time points in the evaluation data belongs to. Then, the abnormality diagnosis system **100** performs abnormality diagnosis for determining whether an abnormality has occurred in the hydraulic pressure control device **240** to be evaluated based on the clustering result.

Series of Processes Executed by Abnormality Diagnosis System **100**

[0043] The flow of processes for abnormality diagnosis executed by the abnormality diagnosis system **100** will now be described with reference to FIG. **5**. The series of processes illustrated in FIG. **5** is executed by the processing circuitry **110** of the abnormality diagnosis system **100**.

[0044] As shown in FIG. **5**, first, in the process of step **S100**, the processing circuitry **110** measures the hydraulic pressure while operating the hydraulic pressure control device **240** in the predetermined diagnosis pattern. The processing circuitry **110** measures the line pressure PL, the first hydraulic pressure Pp, and the second hydraulic pressure Ps while operating the hydraulic pressure control device **240** in accordance with the predetermined diagnosis pattern. The processing circuitry **110** records the measured time-series data on each hydraulic pressure in the storage **120** together with information on the time from the start of measurement.

[0045] In the process of step **S110**, the processing circuitry **110** analyzes the time-series data of each hydraulic pressure recorded in the storage **120** to generate the data on the feature of each hydraulic pressure as evaluation data. The evaluation data is related to an overshoot amount of the hydraulic pressure, a time constant indicating a response delay of the hydraulic pressure, the amplitude in oscillation of the hydraulic pressure, the frequency of the oscillation of the hydraulic pressure, and a damping coefficient of the oscillation of the hydraulic pressure. The processing circuitry **110** records the generated evaluation data in the storage **120**. The evaluation data is data on the features for each time from the start of measurement.

[0046] In the process of step **S120**, the processing circuitry **110** compares the evaluation data with the reference cluster data to cluster the evaluation data. Specifically, the processing circuitry **110** compares the features at multiple time points included in the evaluation data with the cluster data, thereby determining which cluster in the cluster data each of the features at the time points in the evaluation data belongs to.

[0047] In FIG. **4**, an overshoot amount P_trg in the evaluation data is indicated by a black triangle symbol. The processing circuitry **110** calculates, for example, the distance between the overshoot amount in the evaluation data and the centroid of each cluster. Then, the processing circuitry **110** determines that the overshoot amount in the evaluation data belongs to the cluster having the centroid at the closest distance. In the example illustrated in FIG. **4**, the centroid closest to the overshoot amount P_trg in the evaluation data is the centroid of the cluster C_1. As a result, it is determined that the overshoot amount P_trg in the evaluation data belongs to the cluster C_1.

[0048] The clustering method is not limited to such a method of calculating the distance from the centroid. For example, the decision boundary for determining which cluster the feature in the evaluation data belongs to may be determined using a machine learning algorithm such as a support vector machine. The broken lines shown in FIG. **4** are examples of decision boundaries. In the example illustrated in FIG. **4**, the overshoot amount P_trg in the evaluation data is located closer to the cluster C_1 than the decision boundary between the cluster C_1 and the cluster C_3. As a result, it is determined that the overshoot amount P_trg in the evaluation data belongs to the cluster C_1.

[0049] As described above, in step **S120**, the processing circuitry **110** determines which cluster in the cluster data each of the features at multiple time points in the evaluation data belongs to.

[0050] In the process of step **S130**, the processing circuitry **110** stores the clustering result in the storage **120**.

[0051] As illustrated in FIG. **6**, the data on the clustering result stored in the storage **120** is information indicating which cluster the features of the hydraulic pressures are determined to belong to at the time points from the start of measurement. In FIG. **6**, the portion that is determined to belong to the cluster C_1 is indicated by 1, the portion that is determined to belong to the cluster C_3 is indicated by 3, and the portion that cannot be determined to belong to any cluster is indicated by “-”. When the hydraulic pressure is in an equilibrium state, all the features are 0. Therefore, during the period in which the hydraulic pressure is in the equilibrium state in the diagnostic pattern, the processing circuitry **110** cannot determine which cluster the features of the

evaluation data belong to. When the clusters in the cluster data overlap each other, the processing circuitry **110** cannot determine which cluster the features of the evaluation data belongs to. Although not shown in FIG. 6, the portion determined to belong to the cluster C_0 is indicated by 0, the portion determined to belong to the cluster C_2 is indicated by 2, and the portion determined to belong to the cluster C_4 is indicated by 4.

[0052] In the process of step S140, the processing circuitry **110** executes an abnormality determination process. In the abnormality determination process, the processing circuitry **110** determines whether an abnormality has occurred in the hydraulic pressure control device **240** to be evaluated with reference to the clustering result. For example, in a case where the number of times the features are determined to belong to the cluster C_0 is larger than the number of times of times the features are determined to belong to any other cluster, the processing circuitry **110** determines that no abnormality has occurred. For example, the processing circuitry **110** determines that an abnormality corresponding to the cluster, among the clusters C_1 to C_4, to which the features of the evaluation data are determined to belong the largest number of times has occurred.

[0053] In the predetermined diagnosis pattern, a time is divided into periods in accordance with the type of abnormality to be determined, and in each time period, the hydraulic pressures at multiple parts controlled by the hydraulic pressure control device **240** are controlled in a pattern suitable for determining the corresponding abnormality. Therefore, the abnormality determination process may be a process of determining the presence or absence of an abnormality and the type of abnormality by weighting each type of abnormality such that the result of clustering in a period suitable for determining the abnormality is particularly emphasized. The abnormality determination process may be a process using a pre-trained neural network that has been trained in advance through supervised learning so as to determine which abnormality has occurred by inputting data on the clustering result.

[0054] In the process of step S150, the processing circuitry **110** outputs the determination result diagnosed through the abnormality determination process. Specifically, the processing circuitry **110** displays the determination result on the display **130**.

Operation of Present Embodiment

[0055] The processing circuitry **110** of the abnormality diagnosis system **100** generates, as evaluation data, data on the features in hydraulic pressures that are recorded while operating the hydraulic pressure control device **240** to be evaluated in a diagnosis pattern (step S110). The processing circuitry **110** performs clustering to determine which cluster in the cluster data each of the features at multiple time points in the evaluation data belongs to (step S120). The processing circuitry **110** determines whether an abnormality has occurred in the hydraulic pressure control device **240** based on the clustering result (step S140).

[0056] In a case in which an abnormality has occurred in the hydraulic pressure control device **240**, it indicates a hydraulic pressure fluctuation that is different from the hydraulic pressure fluctuation in a normal hydraulic pressure control device **240**. Thus, when the hydraulic pressure control device **240** is operated in the predetermined diagnosis pattern, the data on features indicate changes corresponding to the abnormality status and the type of abnormality. The abnormality diagnosis system **100** determines which cluster each of the features at multiple time points in the evaluation data belongs to through comparison with the cluster data. The abnormality diagnosis system **100** determines whether an abnormality has occurred in the hydraulic pressure control device **240** based on the determination result.

Advantages of Present Embodiment

[0057] (1) The abnormality diagnosis system **100** performs a determination using the data on the fluctuation of the hydraulic pressure of the hydraulic pressure control device **240**. This allows the abnormality diagnosis system **100** to acquire changes in the behavior of a hydraulic pressure resulting from an abnormality and determine that an abnormality has occurred in the hydraulic pressure control device **240**. [0058] (2) The processing circuitry **110** outputs, when determining

that an abnormality has occurred, information on a type of the abnormality that has occurred. This allows the abnormality diagnosis system **100** to identify the type of an abnormality that has occurred. [0059] (3) The features include an overshoot amount of the hydraulic pressure, a time constant indicating a response delay of the hydraulic pressure, the amplitude in oscillation of the hydraulic pressure, the frequency of the oscillation of the hydraulic pressure, and a damping coefficient of the oscillation of the hydraulic pressure. This allows the abnormality diagnosis system **100** to identify the abnormality status and the type of abnormality by acquiring the changes in how the hydraulic pressure fluctuates. [0060] (4) In the diagnosis pattern, a time is divided into periods in accordance with the type of abnormality to be determined, and in each time period, the hydraulic pressures at multiple parts controlled by the hydraulic pressure control device **240** are controlled in a pattern suitable for determining the corresponding abnormality. The abnormality diagnosis system **100** controls the hydraulic pressures at multiples parts in a pattern suitable for determining each abnormality. This allows the abnormality diagnosis system **100** to identify the type of an abnormality more easily. [0061] (5) The hydraulic pressures at the parts controlled by the hydraulic pressure control device **240** include the first hydraulic pressure P_p , which is a hydraulic pressure at the first pulley **210** of a continuously variable transmission, and the second hydraulic pressure P_s , which is a hydraulic pressure at the second pulley **220** of the continuously variable transmission. Thus allows the abnormality diagnosis system **100** to diagnose the hydraulic pressure control device **240** that controls the hydraulic pressure supplied to the first pulley **210** and the second pulley **220**.

Modifications

[0062] The present embodiment may be modified as follows. The present embodiment and the following modifications can be combined as long as they remain technically consistent with each other.

[0063] The abnormality diagnosis system **100** does not have to determine the type of abnormality. For example, the abnormality status and the data on the clustering result shown in FIG. **6** are output. This allows the abnormality diagnosis system **100** to provide information serving as a material for determining not only whether an abnormality has occurred but also the type of abnormality that has occurred. This information allows the operator to identify the type of an abnormality that has occurred.

[0064] The above-described abnormality diagnosis system **100** is configured to determine whether an abnormality has occurred in the hydraulic pressure control device **240**, which controls a hydraulic pressure supplied to the first pulley **210** and the second pulley **220** in a continuously variable transmission. The hydraulic pressure control device **240** to be diagnosed is not limited to the hydraulic pressure control device **240** of the continuously variable transmission. The diagnosis target may be a hydraulic pressure control device of a stepped transmission. The diagnosis target is not limited to the hydraulic pressure control device of the automatic transmission.

[0065] In the above-described embodiment, the abnormality diagnosis system **100** includes the processing circuitry **110** and the storage **120**, and executes software processing. However, this is merely exemplary. For example, the abnormality diagnosis system **100** may include a dedicated hardware circuit (such as ASIC) that executes at least part of the software processes executed in the above-described embodiment. That is, the abnormality diagnosis system **100** may have any one of the following configurations (A) to (C). (A) The abnormality diagnosis system **100** includes an execution device that executes all the processes in accordance with a program and a storage that stores the program. That is, the abnormality diagnosis system **100** includes a software execution device. (B) The abnormality diagnosis system **100** includes an execution device that executes some of the processes in accordance with a program, and a storage. Further, the abnormality diagnosis system **100** includes a dedicated hardware circuit that performs the remaining processes. (C) The abnormality diagnosis system **100** includes a dedicated hardware circuit that executes all the processes. There may be multiple software execution devices and/or dedicated hardware circuits.

That is, the above processes may be executed by processing circuitry that includes at least one of a set of one or more software execution devices and a set of one or more dedicated hardware circuits. The storage (i.e., computer-readable medium) that stores a program includes any type of media that are accessible by general-purpose computers and dedicated computers.

[0066] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. An abnormality diagnosis system for a hydraulic pressure control device, the abnormality diagnosis system comprising: processing circuitry; and a storage, wherein the storage stores cluster data, the cluster data indicating a distribution of clusters data by type of abnormality for features at multiple time points, the cluster data being generated by acquiring, a number of times, data on the features for fluctuation of hydraulic pressure obtained when operating, in a predetermined diagnosis pattern, a hydraulic pressure control device with an identified abnormality status and an identified type of abnormality, and the processing circuitry is configured to: generate, as evaluation data, data on the features in hydraulic pressure that is recorded while operating the hydraulic pressure control device to be evaluated in the diagnosis pattern; and determine whether an abnormality has occurred in the hydraulic pressure control device based on a result of determining which cluster in the cluster data each of the features at multiple time points in the evaluation data belongs to by comparing the features at the time points included in the evaluation data with the cluster data.
 2. The abnormality diagnosis system according to claim 1, wherein the processing circuitry is configured to output, when determining that an abnormality has occurred, information on a type of the abnormality that has occurred.
 3. The abnormality diagnosis system according to claim 1, wherein the features include an overshoot amount of the hydraulic pressure, a time constant indicating a response delay of the hydraulic pressure, an amplitude in oscillation of the hydraulic pressure, a frequency of the oscillation of the hydraulic pressure, and a damping coefficient of the oscillation of the hydraulic pressure.
 4. The abnormality diagnosis system according to claim 2, wherein the diagnosis pattern divides a time into periods in accordance with a type of abnormality to be determined, and control, in each of the periods, hydraulic pressures at multiple parts controlled by the hydraulic pressure control device in a pattern suitable for determining a corresponding abnormality.
 5. The abnormality diagnosis system according to claim 4, wherein the hydraulic pressures at the parts controlled by the hydraulic pressure control device include a first hydraulic pressure and a second hydraulic pressure, the first hydraulic pressure being a hydraulic pressure at a first pulley of a continuously variable transmission, the second hydraulic pressure being a hydraulic pressure at a second pulley of the continuously variable transmission, and the abnormality diagnosis system is configured to diagnose the hydraulic pressure control device that controls the hydraulic pressure supplied to the first pulley and the second pulley.
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